

Speech Transcription Guidelines for Speech Accessibility Project

Phase 3

Target etiologies: Deaf and Hard of Hearing Speech (DHH) and Stuttering Speech¹

0. Introduction

- The transcription is verbatim. All words you hear, including false starts, filler words, and repetitions in normal disfluencies, should be typed as they are. The exceptions are stuttering-like repetitions, which will be transcribed using intelligent verbatim. Do not paraphrase, nor correct grammatical errors. Details are below.
- Standard American written spelling, capitalization, and punctuation should be used.
- No segmentation will be done.

1. Capitalization

- 1.1. The beginning of sentences and proper names should be capitalized.
 - e.g., *I've been reading a book about Sidney Poitier, a great actor.*
 - e.g., *Morning is coffee.*
- 1.2. Spoken letters and acronyms should be capitalized. See 3.1 – 3.4 for more details.

2. Punctuation

- 2.1. Use periods, question marks, exclamation marks, commas, and quotation marks following the conventional standards.
- 2.2. The end of utterances is marked by periods, question marks, or exclamation marks.
 - e.g., *He said, "Are you working, or are you fishing?"*
 - e.g., *Mix that up in there as well and voila!*
 - e.g., *Do I like to travel? Of course!*
- 2.3. Commas are used for listing and especially when it would be confusing to read this part of the transcript without a comma.
 - e.g., *Then, you put mustard, then ham, then salami.*

3. Individual Spoken Letters and Acronyms

- 3.1. Individual spoken letters should be transcribed in caps and preceded by a tilde symbol '~'.
 - e.g., *One of my favorite pets is named Jessye. We spelled it ~J ~E ~S ~S ~Y ~E, after the opera singer.*
 - e.g., *A pangolin is my favorite animal. ~P ~A ~N ~G ~O ~L ~I ~N*
- 3.2. Acronyms that are pronounced as a series of letters should be preceded by '~'.
 - e.g., *My favorite sandwich is a ~BLT.*
 - e.g., *specifically ~WHYY, Philadelphia's ~NPR station*
- 3.3. Acronyms that are pronounced as words should be written with no special markup.

¹ The same standards as those for the Phase 2 etiologies (Parkinson's, amyotrophic lateral sclerosis, cerebral palsy, Down syndrome, stroke) apply, with any changes updated for Phase 3-specific etiologies.

- e.g., *ASAP* (when pronounced ['eɪ.sæp])
- e.g., *HIPAA* (when pronounced ['hɪp.ə])

3.4. When acronyms that are commonly pronounced as a series of letters are spoken as words, only the first letter is capitalized in the transcription to avoid ambiguity in pronunciation.

- e.g., *Ada compliant?* (instead of *ADA compliant?*)

4. Numbers

4.1. All numerals must be written out as complete words according to how they are spoken.

4.2. Hyphenate numbers between twenty-one and ninety-nine

- e.g., *My favorite television show was The Dark Shadows back in the sixties.*
- e.g., *Put all of his money on red twenty-three*

4.3. Stuttering speech: When the Zoom captions have correctly transcribed the numbers as numerals, there is no need to write them out as words.

5. Contractions

5.1. Transcribe contractions such as 'I'm', 'You're', and 'gonna' according to standard spelling.

- e.g., *I'm staying up later at night.*
- e.g., *You're in a safe place*
- e.g., *This is gonna be difficult.*

6. Abbreviations

6.1. Abbreviations should be avoided and words should be transcribed exactly as spoken. However, when abbreviations are used as part of a personal title, they should be transcribed as abbreviations.

- e.g., *Dr. Michael Olken is one of the authors of it.*
- e.g., *I just saw the film again Mr. and Mrs. Smit.*

7. Disfluencies

7.1. Disfluencies are all marked using parentheses.

7.2. **False starts:** sentences and words that are cut short before completion.

7.2.1. They should be transcribed as spoken. They should not be omitted. Mark them using parentheses.

- e.g., *(It's in the) It takes place in the future.*
- e.g., *but (um I just get) the time gets away from me*
- e.g., *The main thing is my partner. (I like to) If I can go with my wife, I like it a lot more.*

7.2.2. When participants have cognitive or reading challenges, speech is transcribed as spoken but without marking false starts in parentheses, because false starts and repairs are difficult to reliably distinguish.

- e.g., *If animals could talk, can where did do you like you like to say?*
- e.g., *Live your could live ~A in world, where would live, world in the world live.*

7.3. Repetitions of full-words or phrases: They should be transcribed and marked using parentheses.

- e.g., *(And) And we would go on lots of trips*
- e.g., *what Boston looked like (back) back then*
- e.g., *(What's the what's the) What's the weather like today?*

7.4. Partial words, either a single sound or syllables, due to repetitions or repairs

7.4.1. When a participant breaks off in the middle of the word, whether that results from repetitions or repairs, partial words should be transcribed using parentheses and a single hyphen '-', without spacing between the hyphen and the word. The hyphen indicates the point at which the word was broken off.

- e.g., *My favorite (s-) sandwich is made by toasting bread*
- e.g., *I'll choose the best of the (a- a-) available choices*
- e.g., *canned (tur-) turkey*

7.4.2. When a single sound is repeated, a letter that represents the sound should be transcribed, instead of the first letter of the spelling, because the first letter can have more than one pronunciation.

- e.g., *from a (k-) concert of the Kingston Trio.*
 - Note. (k-), instead of (c-), is used even though it's a partial word of 'concert' since the letter 'c' is ambiguous with two possible pronunciations (i.e., [k] as in 'cat' or [s] as in 'city').
- e.g., *I like (sh- sh-) Chicago.*
- e.g., *she'll go the (h-) whole way with me*

7.5. Filler words

7.5.1. Participants may use filler words such as 'um, uh, ah' to indicate hesitation or to indicate they are thinking of what to say next. All filler words should be transcribed, using the standardized spellings and marked with parentheses.

- e.g., *And (uh) it's a dry humor.*
- e.g., *I value kindness, (uh) good communication skills*

7.5.2. In contrast, when filler words are 'mentioned' – a case first identified when transcribing stuttering speech – they should not be marked as disfluencies (i.e., they should not be enclosed in parentheses). Instead, double quotation marks are used to indicate that the filler words are being mentioned.

- e.g., *or say, "um um um um", like, that is a stutter.*
- e.g., *It was either just, "um um um um um um" or block.*
-

7.6. Stuttering-Specific: Stuttering-Like Disfluencies

When there are 2 or more repetitions of a sound, a word, or a phrase, they are marked with an asterisk '*'. The exact number of repetitions is not marked, following intelligent verbatim convention. Stuttering-repetitions are enclosed in parentheses, just like other disfluencies.

7.6.1. 2 or more repetitions of a segment are marked with a single hyphen followed by an asterisk: the hyphen indicates that the repeated segment is a partial word.

- e.g., (f-*) *Find m-my phone.*
- e.g., especially (wo-*) *women who stutter.*
- e.g., and I am going (t-*) *to be playing 'blank'*

7.6.2. 2 or more repetitions of a full word are marked with an asterisk and parentheses. No hyphen is used.

- e.g., so (l*) *I have been a person*
- e.g., a conversation (with*) *with my mom recently*
- e.g., I ask him (how*) *how*
- e.g., skating (in*) *in the last (o-*) Olympics?*

7.6.3. 2 or more repetitions of a phrase comprising two words are hyphenated and marked with an asterisk. The first word in the phrase is always a full word.

7.6.3.1. When the second word is a full word, there is no need to add a hyphen at the end.

- e.g., (how-to*) *how to talk*
- e.g., (what-were*) *What were, like, some of your favorite pieces*
- e.g., a name (on-the*) *on the page.*

7.6.3.2. When the second word is cut off, making it a partial word, a hyphen is added at the end.

- e.g., (l-re-*) *I really wish*
- e.g., (that-wa-*) *that was a bit jarring for me*

7.6.4. Prolongations are transcribed using (:) only if longer than 0.5 seconds.

- e.g., (s:)stutter
- e.g., (s-*) (s:)stutter
- e.g., because (m:)my partner's name is
- e.g., I currently have, like (th:)three phone calls I'm, (a-*) avoiding.

7.6.5. Blocks are marked as {b:}, when (1) unexpected pauses between or within words are longer than 0.5 seconds and (2) the sound following the pause has a sudden/abrupt onset, often marked by a burst in loudness or a plosive-like release consistent with tension release.

- e.g., I want to {b:} go home
- e.g., And you know I'm a great l{b:}istener.
- e.g., To have lots of p{b:}atience,

7.6.6. Sounds that suggest non-speech physical actions are marked as {pc:}, short for physical concomitant. Unlike blocks, these secondary behaviors typically produce sounds such as a thump or a rustle.

- e.g., you know, they {pc:} make me mad really.
- e.g., I think (um) {pc:} (l-*) I don't know,
- e.g., and I've {pc:} thought about this in a way, like, playing the piano,

7.6.7. Note. { } are used for both blocks and physical concomitants because the judgement is not 100% certain without visual signals – consistent with our convention of using { } for hard-to-understand words (see 10 for reference).

8. Interjections

8.1. Interjections that convey reactions or sudden expressions should be transcribed, using the standardized spellings: e.g., jeez; huh; oh; okay; uh-huh; uh-oh; whoa; whew; yeah. They are NOT marked with parentheses.

- e.g., *Hm that's a funny question.*
- e.g., *whoa Linda Ronstadt will be our first lady!*

9. Personally Identifiable Information (PII): Privacy

9.1. If participants say their name (the first name, or the last name, or the full name), none should be transcribed to protect the privacy of the participant. We consider the first name alone as a potential PII since disease diagnosis is an example of PII that may in combination allow a person to be identified and our participants frequently mention their diagnosis and city names, which we do not remove as PII.

9.2. If participants say the name of family members, none should be transcribed either.

9.3. If participants say their addresses, ZIP codes, phone numbers, and email addresses, none should be transcribed.

9.4. The tag [PII] is used to indicate the space of the PII. In the corresponding audio file, the section should be replaced with a silent segment.

- e.g., "[PII], we'd like to hear from you".
- e.g., *And I feed [PII] and get her her meds.*
- e.g., *together with [PII], our son*

9.5. Stuttering-Speech Specific: Participants' first names should be transcribed, as stuttering commonly occurs when saying their own names. However, to protect participant privacy, only the first name is transcribed even if a participant's full name is spoken; therefore, the portion of the audio corresponding to the last name should be silenced. For children's names, no part of the name – including the first name – should be transcribed, and the corresponding portion of the audio should be silenced.

10. Hard-to-understand sections: Words or Phrases You Don't Understand

If there is a section of speech that is hard to understand, the section should be marked using curly brackets { }. We use three different codes, depending on the degree of uncertainty.

10.1. **{g: WORD}**: We mark our best guess about the word spoken using {g: WORD}.

- e.g., *you {g: loosen} down, {g: pay} more and then off you go*
- e.g., *And she was my {g: idol}.*

10.2. **{w: NUMBER}**: When it's impossible to make a guess what words were spoken, we mark our best guess about how many words were spoken using {w: NUMBER}.

- e.g., *There was a short article and everything was {w: 3}*
- e.g., *I like to travel by {w: 3} {instead of} by plane*

10.3. **{u: }**: When it's impossible to make a guess even about how many words were spoken, we mark it using {u: }.

- e.g., *Replace {u: } very tightly as tight as possible.*

11. Sentences Spoken in a Foreign Language

11.1. When sentences are spoken in a foreign language, {f: } is used, and no annotation is provided for those sentences.

- e.g., *What is one of your favorite animals and why? (Um) {f: }.*
- e.g., *Talk about the strangest thing you ever eaten. {f: }.*

11.2. If a single word is spoken in a language other than English, it is transcribed, as such words are mostly borrowed words from French or Spanish, typically including food names and proper nouns.

- e.g., *One of my favorite foods is croquetas. (um) They are made of flour and milk, nutmeg, and then for example (um) Spanish jamón.*
- e.g., *I was afraid to something called lagartijo. Lagartijo is like a small animal that climbs trees and stuff like that.*

12. Other Extra Voices

12.1. DHH-Specific

12.1.1. If a caregiver assists the participant during recording, the caregiver's voice may also be captured in the recording. The portion of audio corresponding to the caregiver's speech should be silenced, with no annotation made for this extra voice.

12.1.2. For overlapping speech, silence the section and ensure it is not transcribed.

12.1.3. Background voices maybe ignored as long as the participant's voice is louder and clearly audible. Otherwise,

12.2. Stuttering-Specific

12.2.1. The recording consists of a 40-minute-long conversation between a participant and an interviewer, followed by a 20-minute-long reading. The initial Zoom transcript includes both the participant's and the interviewer's speech. However, only the participant's speech is retained in the recording, and thus only the participant's speech should be reviewed and revised as needed.

12.2.2. In rare cases, the interviewer's voice is recorded. When this happens, no silencing is needed. Instead, ensure that timestamps identify the participant's speech, while square brackets [] identify the interviewer's speech (see 13. Timestamps below for details).

13. Timestamps

13.1. **DHH-Specific:** All recordings must be no more than 30 seconds in duration. If a recording exceeds 30 seconds – which typically occurs in spontaneous speech – the transcript must include timestamps to ensure that each portion of speech can be identified within a section that is no more than 30 seconds. Timestamps are provided in the following format, under the heading #ts.

- e.g.,

[Please explain the steps to check out a book from the library.]
#ts
0.000 20.081

First, you need to make sure you have an updated library card. Then, when you go to the library, you pick out a book or look for a book that you want. Find out exactly where it is in the library and then go find it.
0.081 36.150
And then when you find it, take it to the front desk or the check out desk, hand them your library card, and they'll give you the book and you take it home and then make sure you return it by the expiration or the deadline.

- 13.2. **Stuttering-Specific:** Since all recordings are approximately 1 hour in duration, transcripts must be segmented into sections of no more than 30 seconds. In addition, transcripts include the interviewer's speech that is not retained in the recording. Such speech should be marked using square brackets [] to indicate that no corresponding speech exists in the recording. This is consistent with other uses of [], which are used exclusively to contain informational content (e.g., PII, spontaneous speech question prompts).

- e.g.,

WEBVTT

[00:00:02.560 --> 00:00:13.844
Almpower Interview: Cool! So please introduce yourself. Feel free to share as much or as little as you're comfortable with. If there's private information you want us to remove later, just let us know.]

00:00:13.869 --> 00:00:33.009
Participant: Okay. (um h-) Hi, (um) my name is Annabelle. (um) I'm 29 years old, I live in (bro-) Broward County, Florida. (um) I've been (stu-) stuttering ever since I was a small girl. (um) I'm (ve-) very involved in the (e-) ~NSA community since 2018.

00:00:33.010 --> 00:00:46.730
Participant: (um) I am the co-chapter leader for the ~NSA (ma-) Miami chapter. (um) I'm involved in (sev-*) several stuttering organization, because I (r-) really want to make sure (that um)

00:00:47.408 --> 00:00:53.487
Participant: that I am the representation that others (ke-) can see, because (whe-) when I was growing up,

14. Labeling for Spontaneous Speech Prompts

- 14.1. **DHH-Specific:** In the transcript, spontaneous speech prompts are not deleted but marked using square brackets [] for reference purposes. In other words, square brackets hold information such as PII and spontaneous speech questions (i.e., to indicate the type of question the participant is responding to). By definition, there is no associated audio for this bracketed text. Note: If participants first read the question

and then respond to it, the read question should also be transcribed to accurately reflect what was spoken.

- e.g., *[Talk about a news story you remember from childhood.] A news story that I remember from my childhood happened in (uh), I think, November nineteensixty-three.*
- e.g., *[Tell us about a book, movie, or video you have recently enjoyed.] Tell us about the book, movie, or video you have recently enjoyed. I read the book about the history of the church in Thessalonica.*

14.2. **Stuttering-Specific:** Conversational questions are typically spoken by the interviewer and are not retained in the recording. Therefore, the corresponding transcript should be marked using square brackets []: see 13.2 above for details.

15. What's not transcribed.

15.1. For both DHH and Stuttering speech, we do not mark background noises (e.g., people talking, dog barking, fire truck passing by, sounds from TV). We note any notable noise in the Comment section.

15.2. For both DHH and Stuttering speech, we do not mark non-speech sounds such as laughing, groaning, or throat-clearing.

15.3. **DHH-Specific:** We do not mark sound prolongations or pauses.

15.4. Stuttering-Specific

15.4.1. We mark sound prolongations as described in 7.5.2.

15.4.2. We mark pauses that are judged to be stuttering blocks or physical concomitant, as described in 7.5.3. and 7.5.4.

16. Additional guidelines about reading prompts in DHH speech

16.1. In addition to spontaneous speech, there are three types of reading prompts: digital assistant commands, novel sentences, and single words. The main transcription tasks for reading prompts are to verify that participants have correctly read the text prompts, and to edit the transcript to reflect errors (e.g., if a word is omitted) so that the transcript matches what participants actually said. Disfluencies should be marked the same way as in spontaneous speech.

16.2. If participants add spontaneous speech while reading prompts, the spontaneous speech portion should be marked using (ss:), to distinguish it from the read-speech.

- e.g., *Call the nearest coffee shop. (ss: I hope it's not Starbucks).*

16.3. If participants respond to the reading prompt instead of simply reading it, their response should be transcribed the same way as spontaneous speech. We may write 'responded as ss' in the Comment section. Similarly, if participants only read a spontaneous speech prompt without responding to it, what they read should be transcribed, and may write 'only read' in the Comment section.

17. Invalid Audio Files

When audio files are not valid for the following reasons, they are excluded from the database. The following codes are noted in the Comment section to specify the reason for exclusion.

17.1. **nod:** no data.

- This is when there is no sound at all and a .wav file does not even exist.
- This was due to a technical glitch in the very initial stage, which had been fixed in March 23, 2023, and rarely occurred afterwards.

17.2. **nov**: no voice or no relevant voice

- This is when there is no participant's voice at all, or their voice exists but they were not aware of the recording status, making it an accidental recording of conversation.

17.3. **bgn**: background noise.

- This is when the participant's voice is present, but background noise is loud enough to mask the voice.

17.4. **otv**: presence of other voice(s).

- This is when there is a voice, or multiple voices, but none of them are from the participant.

17.5. **cut**: voice cut off.

- This is when the participant's recording is cut off abruptly, leaving less than one word.
- Note. When there is an abrupt cut off, but if at least one full word is produced, we do not exclude the file.

17.6. **loa**: low amplitude

- This is when the intensity of voice is substantially low throughout the recording, generally defined as when amplitude is below 5dB.

17.7. **nrw**: no real word

- This is when the speech production contains only no real words in English, a case often found in participants with aphasia secondary to stroke.

18. SLP comments.

18.1. When SLP comments are added in the comment fields, they are marked using square brackets.

- e.g., [Pace more rapid than would be expected with a digital assistant.]
- e.g., [Creaky voice.]

Summary of the guidelines with examples

Category	Instruction		Examples
Capitalization	The beginning of sentences, proper names, individual spoken letters and acronyms are capitalized		• <i>Sidney Poitier</i> • <i>Morning is coffee.</i>
Individual spoken letters	Capitalized and preceded by a tilde symbol ~		• ~J ~E ~S ~S ~Y ~E • ~P ~A ~N ~G ~O ~L ~I ~N
Acronyms pronounced as a series of letters	Capitalized and preceded by a tilde symbol ~		• ~WHYY, Philadelphia's ~NPR station
Acronyms pronounced as words	Written as such, with no special markup		• HIPAA
Numbers	Written out. Numbers between twenty-one and ninety-nine are hyphenated Note. stuttering		• <i>in the sixties</i> • <i>twenty-three sixties</i>
Contractions	Standard spelling		• <i>I'm staying up later</i> • <i>This is gonna be difficult</i>
Abbreviations	Personal titles are transcribed as abbreviations		• <i>Dr. Michael Olken</i> • <i>Mr. and Mrs. Smith</i>
Disfluencies	False starts are transcribed as spoken, with no special codes except parentheses		• <i>(They could) if they just waited another minute</i> • <i>(I mean there's) Ask him some questions about that.</i>
	Repetitions of full-words or phrases are transcribed as spoken, with no special codes except parentheses		• <i>(back) back then</i> • <i>(What's the what's the) What's the weather like today?</i>
	Partial words are transcribed as spoken, with a single hyphen and parentheses		• <i>My favorite (s-) sandwich</i> • <i>canned (tur-) turkey</i> • <i>I watched every one of the (reserv-) renovation</i>
	Filler words (hesitation sounds) are transcribed using standard spellings and parentheses		• <i>(uh) it's a dry humor.</i> • <i>I value kindness, (uh) good communication skills</i>
	2 or more repetitions of segments are marked with a single hyphen followed by an asterisk. This applies to both DHH and Stuttering speech		• <i>(f-*) Find m-my phone.</i> • <i>What's the (w-*) weather like today?</i>
	Stuttering events: This applies only to Stuttering group.	2 or more repetitions of a full word are marked with an asterisk and parentheses: no hyphen.	• <i>so (l*) I have been a person</i> • <i>a conversation (with*) with my mom recently</i>
		2 or more repetitions of a phrases comprising two words are connected with a hyphen and marked with an asterisk. The first word in the phrase should be a full word. When the second word is a full word, no need to add a hyphen at the end. When the second word is cut off and thus making it a partial word, a hyphen is added at the end.	• <i>(how-to*) how to talk</i> • <i>(what-were-*) What were, like, some of your favorite pieces</i> - vs. • <i>(I- re-*) I really wish</i> • <i>(that-wa-*) that was a bit jarring for me</i>

		Prolongations are transcribed using (:) if longer than 0.5 second. Blocks are marked as {b:} when unexpected pauses between words or within words are longer than 0.5 seconds. Sounds that suggest non-speech physical actions are marked as {pc:}. Unlike blocks, they may produce sounds such as a thump or a rustle.	• (s:)stutter • because (m:)my partner's name is • I want to {b:} go home • To have lots of p{b:}atience, • And you know I'm a great l{b:}istener. • you know, they {pc:} make me mad really.
Interjections	Transcribed using standard spellings. No special marking		• whoa Linda Ronstadt will be our first lady!
Personally Identifiable Information (PII)	The following PII are marked as [PII]: • A participant's first name, last name, full name, and the name of family members. • A participant's addresses, ZIP codes, phone numbers, and email addresses. • In the corresponding audio file, the PII section is replaced with a silent segment. • For stuttering:		• "[PII], we'd like to hear from you". • together with [PII], our son,
Hard-to-understand sections: words or phrases you don't understand	Best guess about the word spoken is marked using {g: WORD}		• she was my {g: idol}.
	When it's impossible to guess what words were spoken, best guess about how many words were spoken is marked using {w: NUMBER}		• There was a short article and everything was {w: 3}
	When it is impossible to make a guess even about how many words were spoken at what was said, the section is marked using {u: }		• Replace {u: } very tightly as tight as possible.
Punctuation	The end of utterances is marked by periods, question marks, exclamation marks. Commas are used for listing and especially when it would be confusing to read this part of the transcript without a comma		• He said, "Are you working, or are you fishing?" • Of course! • Then, you put mustard, then ham, then salami.
Sentence(s) spoken in a foreign language	{f: } is used and no annotation is provided for the		• What is one of your favorite animals and why? (Um) {f: }.
Additional speech during reading	If participants add spontaneous speech while reading prompts, the		• Call the nearest coffee shop. (ss: I hope it's not Starbucks).
Summary of non-letter characters to code specific meanings	~	individually spoken letters	See above examples
	()	all types of disfluency	
	-	partial words when used inside parentheses	
	*	stuttering-repetitions	
	(:)	prolonged sound	
	[]	spontaneous speech prompts, PII, and SLP comments, i.e., no associated audio exists.	
	{ }	hard-to-understand sections: {g:}, {w:}, {u:}	

	stuttering-specific: {b:} for stuttering blocks {pc:} for physical concomitant in stuttering
{f: }	sentence(s) spoken in a foreign language.
(ss:)	spontaneous speech appearing alongside read speech during reading tasks
[]	spontaneous speech prompts, PII, and SLP comments, i.e., no associated audio exists.

Appendix: Revision history

Date	Edited by	Notes
May, 2023	Heejin Kim	V1 for Phase 1; Parkinson's
August, 2024	Heejin Kim	V2 for Phase 2: Parkinson's, amyotrophic lateral sclerosis, cerebral palsy, Down syndrome, stroke
August, 2026	Heejin Kim	V3 for Phase 3: Deaf and Hard of Hearing Speech, Stuttering